Wenyuan Wang

codemaker-123.github.io github.com/codemaker-123
Google Scholar ww462@scarletmail.rutgers.edu

RESEARCH INTERESTS

My research focuses on multimodal learning, world models and interpretability. I aim to develop AI systems capable of robust causal forecasting and grounded understanding of the physical world.

EDUCATION

Wuhan University
B.S. in Electronic Information Engineering — *Sep 2020 - June 2024*
– GPA: 3.60(87.2)/4.0

Research Experience

Rutgers University, New Brunswick — **Advised by Prof. Hao Wang**
Research Assistant — *Mar 2024 – Mar 2025*

University of Illinois Urbana-Champaign — **Advised by Prof. Tong Zhang**
Research Assistant — *Mar 2025 – Present*

University of California, San Diego — **Advised by Prof. Biwei Huang**
Research Assistant — *Sep 2025 – Present*

Publications

- **Generalizable Geometric Image Caption Synthesis**
  Yue Xin*, **Wenyuan Wang***, Rui Pan, Ruida Wang, Howard Meng, Renjie Pi, Shizhe Diao, Tong Zhang
  arXiv:2509.15217/ICLR 2026 (*Under Review*)
  - Integrated symbolic reasoning with neural-guided heuristics, constructed a verifiable pipeline for image-pair generation.
  - Proposed RAFT pipeline to enhance dataset quality and improve geometric problem-solving and generalization capabilities of the model.
- **Probabilistic Residual User Clustering**
  Wenyuan Wang, Yusong Zhao, Zihao Xu, Hengyi Wang, Shreya Venugopal, Desmond Lobo, Chengzhi Mao, Qi Xu, Zhigang Hua, Yan Xie, Bo Long, Shuang Yang, Hao Wang
  IJCAI2025 workshop / Submitted to TMLR
  - Proposed PRUC, a causal Bayesian framework that clusters users and models residuals to enhance recommendation accuracy
  - Introduced plug-and-play architecture compatible with diverse deep learning recommenders, improving cold-start performance
- **PhysProver: Advancing Automatic Theorem Proving for Physics**
  Hanning Zhang, *Ruida Wang*, *Rui Pan*, ***Wenyuan Wang***, BingXu Meng, Tong Zhang
  arXiv preprint arXiv:2601.15737/ARR, ACL2026 (*Submitted to ACL*)
  - Curated dataset containing 3,122 training examples from the PhysLean library, covering topics such as category theory, tensor algebra, and relativity physics. Finetuned Goedel-Prover-V2 on this domain-specific dataset.
  - Proposed an evaluation framework utilizing Lean server verification, enabling scalable and reliable assessment of theorem proving in physics
- **Superstructuring a Task-Sufficient World Model on Top of Visual Foundational Model**
  Minghao Fu, Fan Feng, **Wenyuan Wang**, Biwei Huang
  WMW 2026 Workshop(*Under Review*)
  - Analyze the latent space of the world model using t-SNE clustering visualization and to further interpret our model by visualizing its attention patterns.

- **Multimodal Needle in a Haystack: Benchmarking Long-Context Capability of MLLMs**
Hengyi Wang, Haizhou Shi, Shiwei Tan, Weiyi Qin, **Wenyuan Wang**, Tunyu Zhang, Akshay Nambi, Tanuja Ganu, Hao Wang
NAACL 2025 Main
  - Evaluated performance of InstructBLIP on multimodal long-context benchmark
  - Contributed to comprehensive evaluation methodology for multimodal LLM capabilities
- **Continual Learning of Large Language Models: A Comprehensive Survey**
Haizhou Shi, Zihao Xu, Hengyi Wang, Weiyi Qin, **Wenyuan Wang**, Yibin Wang, Zifeng Wang, Sayna Ebrahimi, Hao Wang
ACM Computing Surveys
  - Researched and documented advancements in continual learning for multimodal LLMs
  - Contributed to comprehensive survey covering state-of-the-art techniques in the field
- **Multi-tailed Vision Transformer for Efficient Inference**
Yunke Wang, Bo Du, **Wenyuan Wang**, Chang Xu
Neural Networks, 2024, 174: 106235
  - Designed multiple tails to generate visual sequences of different lengths for Transformer processing
  - Achieved significant FLOP reduction with no accuracy degradation compared to baseline methods

RESEARCH PROJECTS

World Model-Based Simulator with VLM-based Evaluation ***Advisor: Biwei Huang****, Oct 2025 – Dec 2025*
- Developed a framework using Ctrl-World as a simulator to generate scalable action trajectory rollouts.
- Employed QwenVL3-32B as a vision-language evaluator to assess and filter synthetic data quality.

Physics Diagram Synthesis ***Advisor: Tong Zhang****, Oct 2025 - Present*
- Propose a method for generating synthetic physics problem solutions using reinforcement learning with augmented feedback.
- Leverage MLLM to iteratively improve reasoning and visualization generation through automated evaluation and data selection.

Steering Vector Analysis in Large Reasoning Models ***Advisor: Canyu Chen****, Jun 2025 - Jul 2025*
- Investigated how steering-vector-based interventions affect the robustness and faithfulness of reasoning in large reasoning models.
- Analyzed intrinsic properties of steering vectors and demonstrated how specific latent directions systematically induce, modify, or bias step-by-step reasoning trajectories.

Interpretability of Multimodal Large Language Models ***Advisor: Prof. Hao Wang****, Mar 2024 - Sep 2024*
- Identified strong attention correlations between visual inputs and token-level outputs in LLaVA.
- Proposed an adaptive pruning technique for hierarchical attention layers, improving both accuracy and inference efficiency.

Dynamic Scene Graph Generation ***Advisor: Prof. Runhao Zeng, Prof. Chuang Gan****, Jun 2023 - Oct 2023*
- Deployed a MiniGPT-4–based multimodal LLM for scene graph generation with joint reasoning over class, box, and relationship triples.
- Fine-tuned class and bounding-box models via transfer learning to improve relationship prediction accuracy.

Dynamic Confidence Calibration with Adaptive Temperature ***Advisor: Prof. Chang Xu,*** *Nov 2022 - Jan 2023*
- Proposed a dynamic neural network that allocates computational resources according to sample difficulty.
- Applied adaptive temperature adjustment to improve confidence calibration and sample difficulty verification.

WORK EXPERIENCE

Visual Alignment in Industrial Settings *Aug 2023 – Jan 2024*
Intern, Siemens Shanghai, China
- Simulated AGV-based cargo detection in Gazebo using ROS2.
- Designed a vision algorithm to estimate cargo pose and position by analyzing frontal-area geometry.

SKILLS

- **Programming:** Python, C++, C, Lean, Pytorch
- **Tools & Platforms:** ROS, Linux, Docker, Embedded Systems, Gazebo, IsaacSim